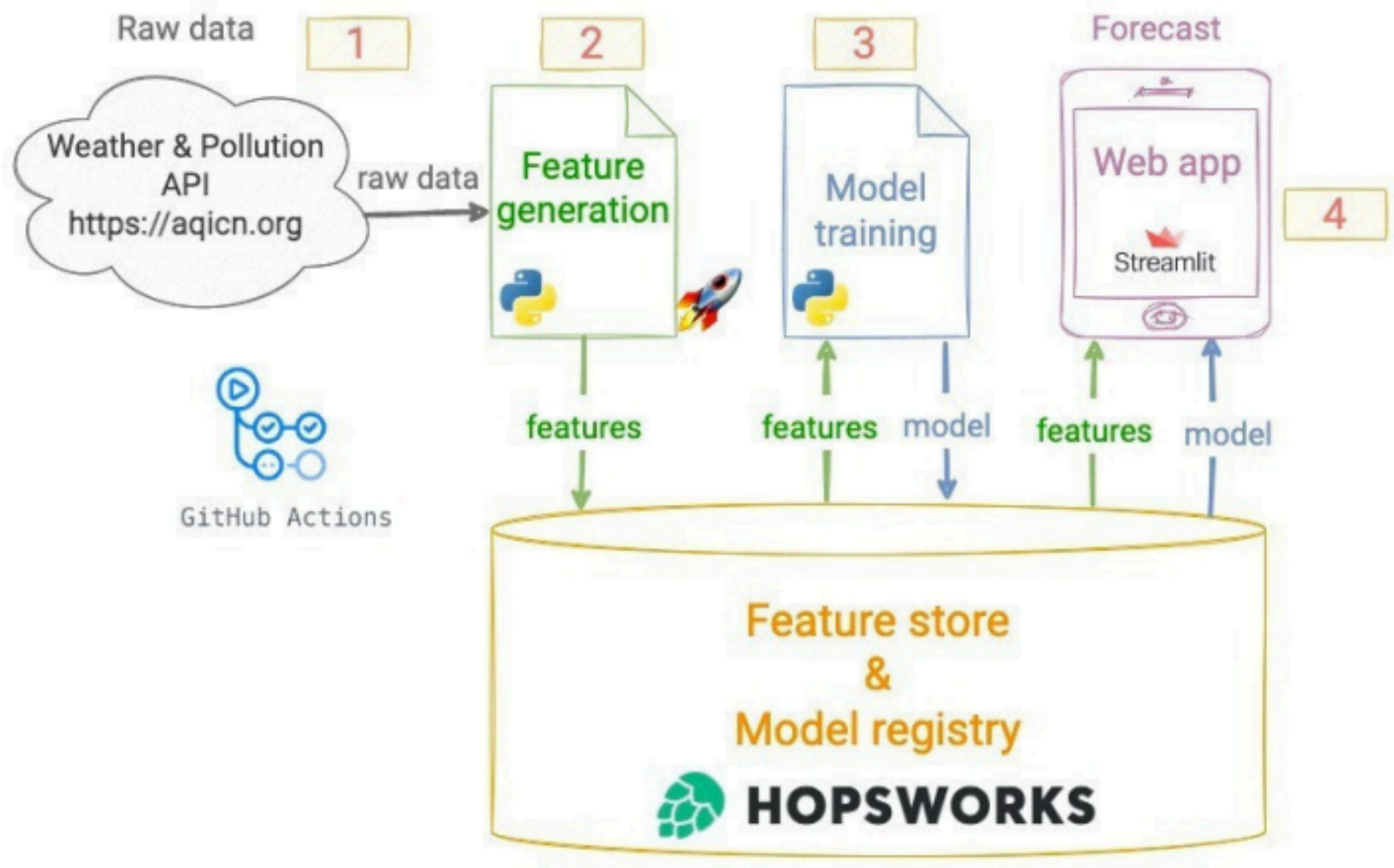


Pearls AQI Predictor

Let's predict the Air Quality Index (AQI) in your city in the next 3 days, using a 100% serverless stack.

The following is a high level overview for you to achieve this

Air Quality Index (AQI) prediction service



Write a Python script that:

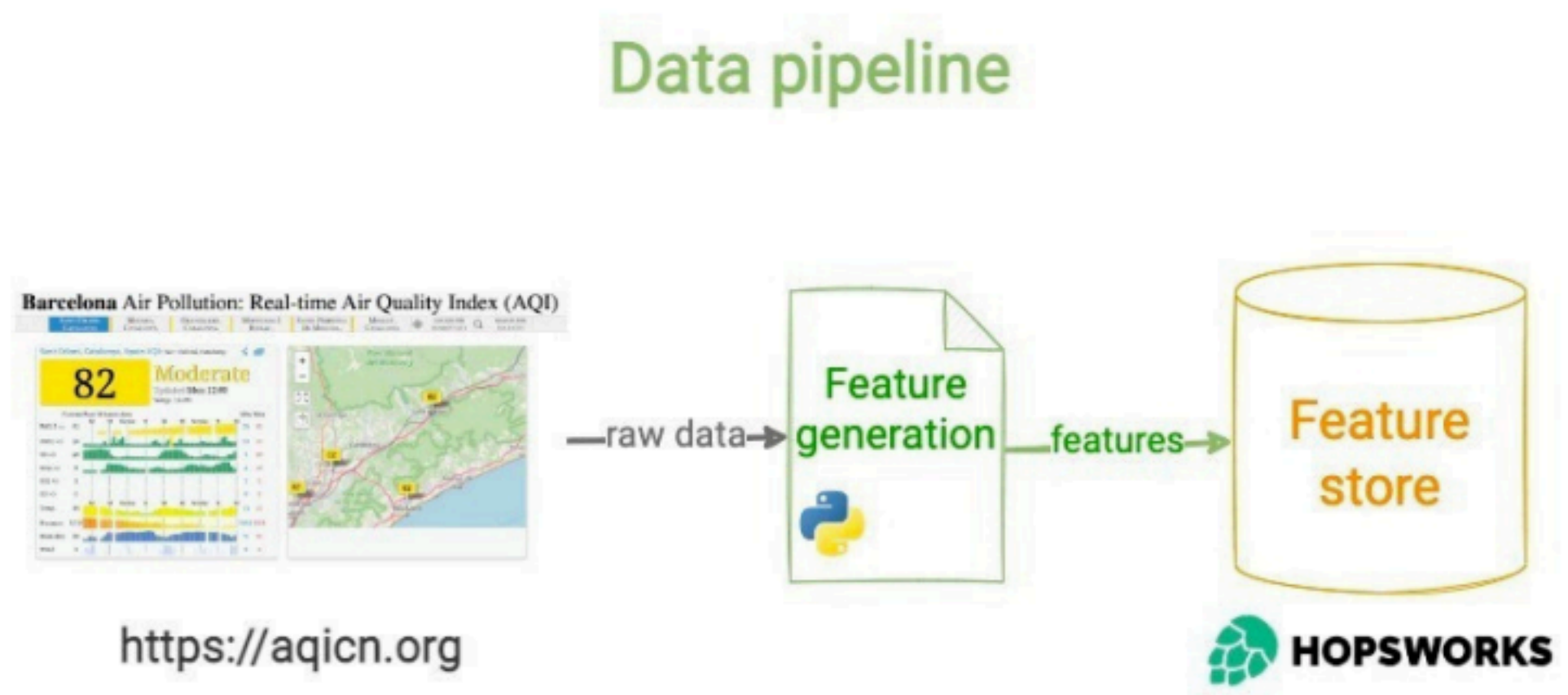
1 → Fetches raw weather and pollutant data from an external API like AQICN or OpenWeather

(The above api is just an example, you may need to explore other options too)

2 → Computes features from this raw data (aka model inputs), and targets (aka model outputs)

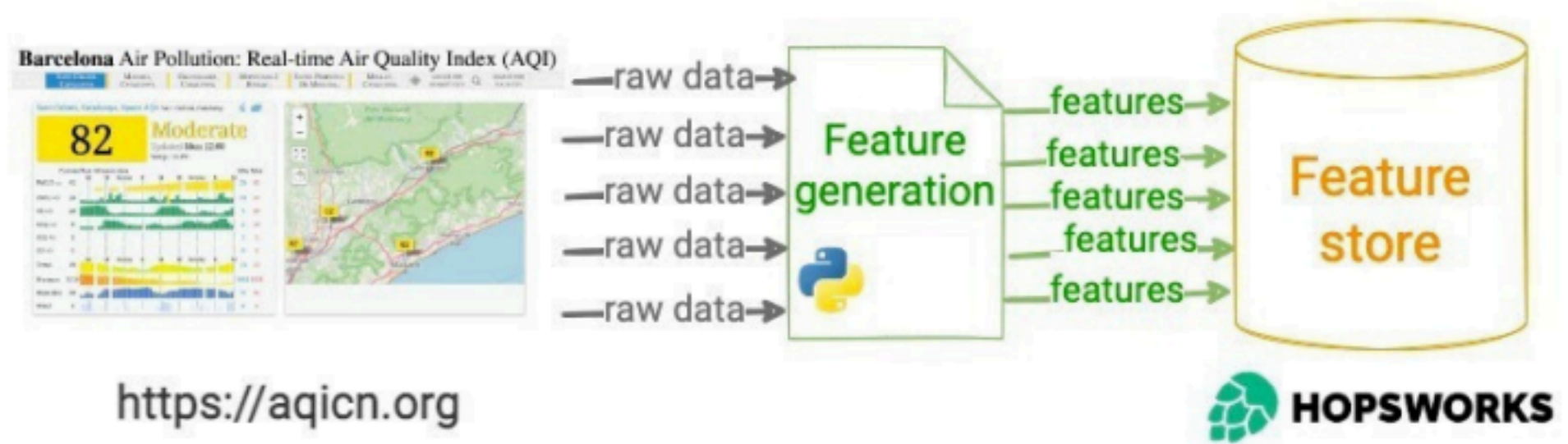
Include time-based features (hour, day, month) and derived features like AQI change rate.

3 → Stores these features in the Feature store You may want to explore Hopsworks or Vertex AI (Free tiers)



Run the feature script from step 1 for a range of past dates, to generate training data for your ML models.

Backfill data



Write a Python script that:

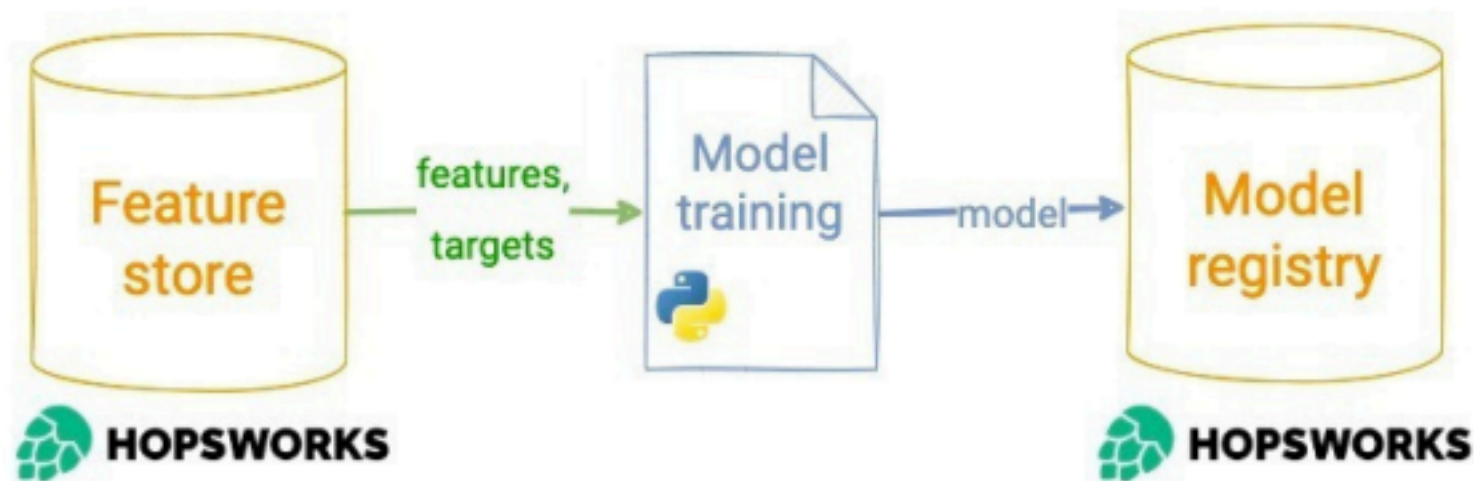
1 → Fetches historical (features, targets) from the Feature Store.

2 → Trains and evaluate the best ML model possible for this data

*Experiment with Scikit-learn models (Random Forest, Ridge Regression) and TensorFlow/PyTorch for advanced models.
Evaluate performance using RMSE, MAE, and R^2 .*

3 → Stores the trained model in the Model Registry.

Model training pipeline



Create a CI/CD pipeline that automatically runs

- the **feature script every hour**, and
- the **training script every day**.

Some popular and free CI/CD tools that you can use are Apache Airflow and Github Actions but you are encouraged to explore other tools too

Data pipeline

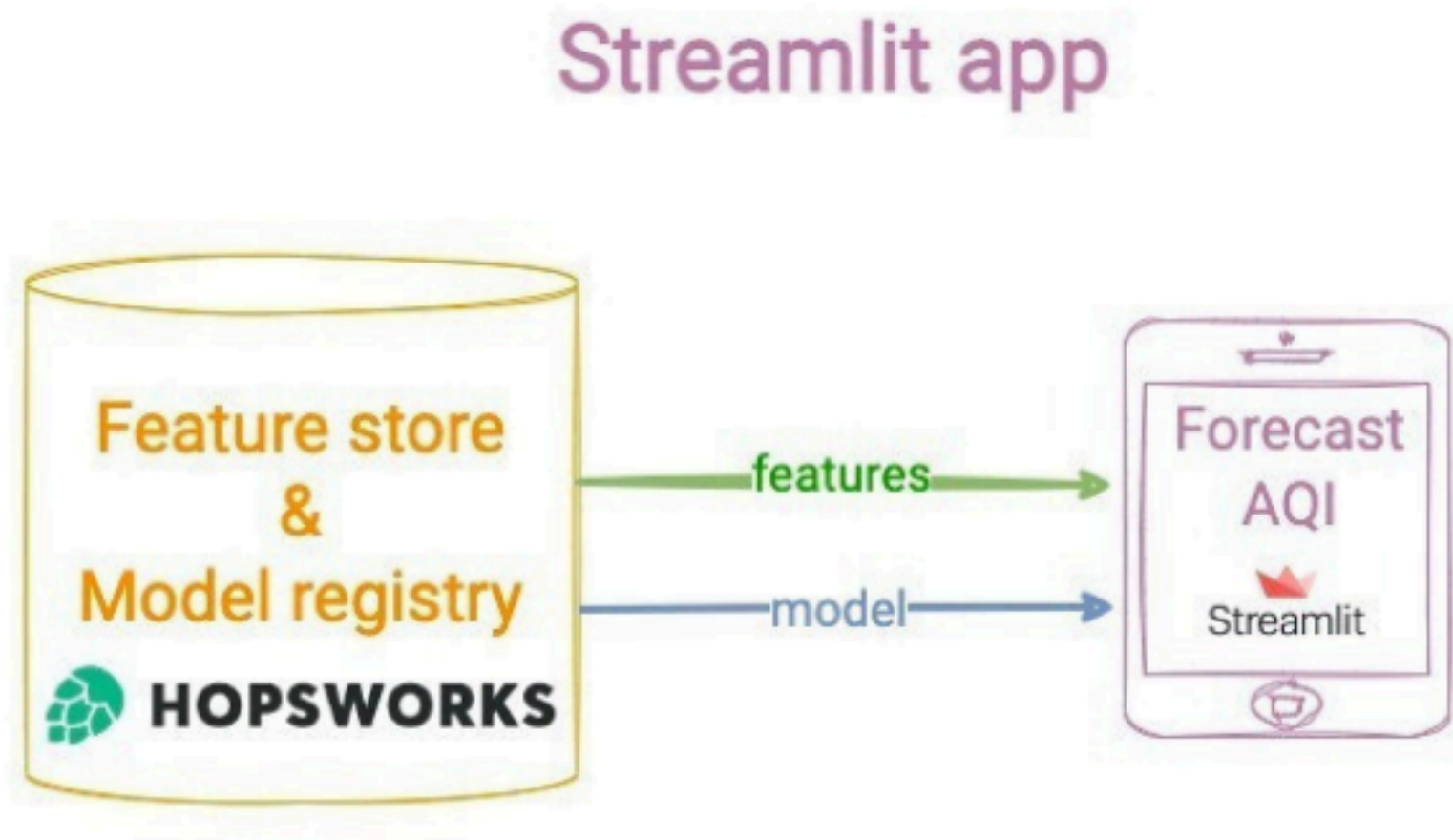
Runs every 1h



Your app

- 1 → Loads the model and features from the *Feature Store*,
- 2 → Computes model predictions and shows them on a simple and descriptive dashboard.
- 3 → Use Streamlit/Gradio and

Flask/FastApi for the web app



Perform EDA to identify trends.

You should use a variety of forecasting models, from statistical modelling to deep learning models

Use SHAP or LIME for feature importance explanations. Add alerts for hazardous AQI levels

Final Submissions:

1. End-to-end AQI prediction system.
2. A scalable, automated pipeline.
3. An interactive dashboard showcasing real-time and

forecasted AQI data.

4. A detailed report documenting everything you managed to achieve